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Abstract

Thermal history is geometrically encoded in the atomic structure of glasses, but whether the geometric carrier of this memory is universal across glass families or system-dependent has not been tested. Here we apply a GATv2 graph neural network structural probepreviously validated on Lennard-Jones (LJ) and Kob–Andersen (KA) pair-potential glasses—to amorphous silica (SiO2), a tetrahedral network glass with fundamentally different bonding. Using published molecular dynamics configurations of 3000-atom glasses at four cooling rates spanning three decades (0.1–100 K/ps), with spatial patch extraction and stratified group k-fold crossvalidation to prevent data leakage, we find that the geometric carrier of thermal history is system-dependent. Bond-length statistics, which achieve >95% classification accuracy in pair-potential glasses, carry no discriminative signal in SiO2 (43.3 ± 8.2%, consistent with chance). Adding bond-angle features (O–Si–O tetrahedral and Si–O–Si bridging angles) restores accuracy to 96.3 ± 5.2% (AUC = 0.981) under five-fold group cross-validation. Simultaneously, Kolmogorov–Smirnov tests on all three partial pair distribution functions (Si–O, Si–Si, O–O) show no statistically significant difference between the fastest and slowest glasses ($p > 0.05$ for all pairs), establishing that the dominant structural carrier of thermal memory in silica is invisible to standard diffractionbased characterisation. These results demonstrate that the GATv2 architecture transfers without modification across glass families, but discovers qualitatively different geometric carriers in each: bond-length distributions in pair-potential glasses versus bond-angle correlations in network glasses.

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System-Dependent Geometric Carriers of Thermal History in Glass: Angular Geometry Encodes Cooling-Rate Memory in Amorphous Silica Where Bond Lengths and Pair Distribution Functions Cannot

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Abstract

Thermal history is geometrically encoded in the atomic structure of glasses, but whether the geometric carrier of this memory is universal across glass families or system-dependent has not been tested. Here we apply a GATv2 graph neural network structural probe—previously validated on Lennard-Jones (LJ) and Kob–Andersen (KA) pair-potential glasses—to amorphous silica (SiO_2), a tetrahedral network glass with fundamentally different bonding. Using published molecular dynamics configurations of 3000-atom glasses at four cooling rates spanning three decades (0.1–100 K/ps), with spatial patch extraction and stratified group k-fold cross-validation to prevent data leakage, we find that the geometric carrier of thermal history is *system-dependent*. Bond-length statistics, which achieve >95% classification accuracy in pair-potential glasses, carry no discriminative signal in SiO_2 ($43.3 \pm 8.2\%$, consistent with chance). Adding bond-angle features (O–Si–O tetrahedral and Si–O–Si bridging angles) restores accuracy to $96.3 \pm 5.2\%$ (AUC = 0.981) under five-fold group cross-validation. Simultaneously, Kolmogorov–Smirnov tests on all three partial pair distribution functions (Si–O, Si–Si, O–O) show no statistically significant difference between the fastest and slowest glasses ($p > 0.05$ for all pairs), establishing that the dominant structural carrier of thermal memory in silica is invisible to standard diffraction-based characterisation. These results demonstrate that the GATv2 architecture transfers without modification across glass families, but discovers qualitatively different geometric carriers in each: bond-length distributions in pair-potential glasses versus bond-angle correlations in network glasses.

1 Introduction

The glass transition encodes the thermal history of a material into its atomic structure [1, 2]. Glasses produced at different cooling rates occupy different basins of the potential energy landscape [3], leading to measurable differences in density, stiffness, and transport properties even when chemical composition is identical [4]. Identifying *where* in the atomic geometry this memory resides is a fundamental question in glass physics with direct implications for materials characterisation and process control.

Recent work demonstrated that graph neural networks (GNNs) can serve as structural probes for thermal history in model glasses [5]. In single-component Lennard-Jones (LJ) and Kob–Andersen (KA) binary glasses, a GATv2 architecture [6] trained on local bond-length statistics achieves >95% accuracy in classifying fast-cooled versus slow-cooled configurations from single structural snapshots [5]. Systematic ablation experiments established that the discriminative signal resides in the *metric geometry* of the bond network—specifically, the statistical distribution of local bond lengths and their spatial organisation—rather than in graph topology or global scalar properties [5]. Subsequent studies extended this framework to cyclic mechanical fatigue [7], competing thermal and mechanical histories [8], and latent memory recovery [9], demonstrating that the GATv2 structural probe captures physically meaningful information about the potential energy landscape.

However, all prior validation has been conducted on *pair-potential* glass formers: the single-component LJ system and the KA binary mixture, both of which employ isotropic, non-directional interatomic potentials. Real glasses of technological importance—silicates, phosphates, chalcogenides—are *network glasses* in which directional covalent bonds form a connected topology of corner-sharing polyhedra [10, 11]. In amorphous SiO_2, the prototypical network glass, each Si atom is tetrahedrally coordinated by four O atoms, and each O bridges two Si atoms. The resulting structure is characterised not only by bond lengths but also by bond angles (O–Si–O tetrahedral angles $\approx 109.5°$, Si–O–Si bridging angles $\approx$ 130–180°) and ring topology [12, 11].

This structural richness raises a question that the pair-potential studies cannot address: *Is the geometric carrier of thermal history universal across glass families, or does it depend on the bonding topology of the system?* Specifically, does thermal history in network glasses encode in

bond-length distributions (as in LJ/KA), in bond-angle distributions (unique to network formers), or in some combination?

Here we address this question by applying the identical GATv2 framework from Refs. [5, 7, 8, 9] to amorphous SiO_2, using published molecular dynamics configurations from the DM2 dataset [16]. The key methodological choice is to change *nothing* about the GNN architecture, training protocol, or cross-validation strategy—only the input features are varied through systematic ablation—so that any differences in which features carry the discriminative signal can be attributed to the physics of the glass, not to modelling choices.

2 Methods

2.1 Dataset

We use amorphous SiO_2 configurations from the publicly available DM2 dataset [16], generated by melt-quench molecular dynamics using the Jakse interatomic potential [17] in LAMMPS [18]. The dataset contains 3000-atom configurations at four cooling rates: 0.1, 1, 10, and 100 K/ps, with six independent realisations at each rate (24 structures total). The cooling rates span three decades, providing a 1000× contrast between the slowest and fastest glasses. Short-range interactions are computed with an 8.0 Å cutoff, and Coulomb interactions use the Fennell damped shifted force model with a damping parameter of 0.25 $Å^{-1}$ and a global cutoff of 8.0 Å, as described in Ref. [16].

For binary classification, we use the two extreme cooling rates: 0.1 K/ps (slow, label 1) and 100 K/ps (fast, label 0), yielding 12 structures (6 per class). This is a small dataset by machine learning standards; we compensate through spatial patch extraction and group-aware cross-validation.

2.2 Spatial patch extraction

Each 3000-atom structure is decomposed into $n_p = 10$ spatial patches of $N_{\text{sub}} = 512$ atoms, following the protocol established for the KA system in Ref. [5]. For each patch, a random centre atom is selected and its 512 nearest neighbours (under periodic boundary conditions with minimum image convention) are retained. This yields $12 \times 10 = 120$ graphs for binary classification (60 per class). All patches from a given parent structure share a group label to prevent information leakage during cross-validation.

2.3 Graph construction

Each patch is represented as an attributed graph $G = (V, E, \mathbf{X}, \mathbf{e})$. Vertices $V = \{1, \ldots, N_{\text{sub}}\}$ correspond to atoms. Edges connect all atom pairs within a cutoff radius r_c. Edge features are the raw interatomic distances $e_{ij} = \|\mathbf{r}_j - \mathbf{r}_i\|$ under minimum image convention.

The choice of cutoff is critical for network glasses. Table 1 quantifies the graph structure as a function of r_c for a representative SiO_2 patch. At $r_c = 2.0$ Å, only Si–O first-neighbour bonds (∼1.62 Å) are captured. Every Si has exactly four O neighbours and every O has exactly two Si neighbours, producing graphs with *identical topology for every glass* regardless of thermal history. There are zero Si–Si and zero O–O edges. This explains why any GNN with short cutoff cannot distinguish cooling rates: it receives structurally indistinguishable inputs.

We use $r_c = 5.0$ Å, which captures Si–O (∼1.62 Å), O–O (∼2.65 Å), and Si–Si (∼3.12 Å) correlations as well as second-shell structure. At this cutoff, each atom has ∼25 neighbours and the graph contains ∼13 000 edges, providing the GNN with access to medium-range structural information.

Table 1: Graph statistics as a function of cutoff radius for a 512-atom SiO_2 patch. At $r_c = 2.0$ Å, only Si–O bonds are captured, yielding identical graph topology for all glasses.

r_c (Å)	Edges	Neigh/atom	Si–O	Si–Si	O–O
2.0	512	2.7	512	0	0
3.0	1378	7.2	520	76	782
3.5	2068	10.8	782	258	1028
4.0	3344	17.4	1620	276	1448
5.0	7522	39.2	3406	798	3318
6.0	13152	68.5	5810	1506	5836

2.4 Node feature sets

We define four feature sets for systematic ablation:

Experiment A: Bond-only (5D). Per-atom features are $\mathbf{x}_i = [\bar{r}_i, \sigma_{r,i}, r_i^{\min}, r_i^{\max}, \tilde{d}_i]$, where $\bar{r}_i$ is the mean distance to all neighbours within r_c, $\sigma_{r,i}$ its standard deviation, $r_i^{\min}$ and $r_i^{\max}$ the extremal distances, and $\tilde{d}_i = d_i/d_{\max}$ the normalised coordination number. These are identical to the features used for LJ and KA systems in Ref. [5].

Experiment B: Bond + Angle (9D). The 5D bond features are augmented with four angular features computed from first-shell neighbours within $r_c^{\text{angle}} = 2.0$ Å: for Si atoms, the mean and standard deviation of all O–Si–O bond angles formed by coordinated oxygens; for O atoms, the Si–O–Si bridging angle (when bridging two Si atoms). The total per-atom feature vector is $\mathbf{x}_i \in \mathbb{R}^9$.

Experiment F: Bond (5D) + instance normalisation. The 5D bond features are z-score normalised per graph: $\mathbf{x}_i \leftarrow (\mathbf{x}_i - \boldsymbol{\mu}^{(s)})/(\boldsymbol{\sigma}^{(s)} + \epsilon)$, where $\boldsymbol{\mu}^{(s)}$ and $\boldsymbol{\sigma}^{(s)}$ are computed across all atoms in graph s. This removes global magnitude information (e.g., average bond length, density), forcing the model to rely on relational geometry [5].

Experiment G: Extended bond (8D). The 5D features are extended with a skewness proxy for the local distance distribution, the 25th percentile, and the 75th percentile, matching the format used in Refs. [7, 8, 9].

2.5 GATv2 architecture

The GATv2 classifier is architecturally identical to all previous studies in this series [5, 7, 8, 9]. It consists of: (i) a linear encoder mapping input features to 64 dimensions; (ii) two GATv2 message-passing layers [6] with 4 attention heads and edge dimension 1, with concatenated multi-head output; (iii) global mean pooling over all nodes; (iv) a post-pooling MLP ($256 \to 128 \to 64$, with LayerNorm, ReLU, and Dropout with $p = 0.2$); and (v) a single-logit binary classifier trained with BCEWithLogitsLoss. No architectural hyperparameters were modified from the pair-potential studies.

2.6 Training and evaluation

All experiments use five-fold stratified group k-fold cross-validation (StratifiedGroupKFold from scikit-learn [19]). The group variable ensures that all patches from the same parent 3000-atom structure are assigned to the same fold, preventing data leakage that would inflate accuracy estimates. With 12 structures (6 per class), each fold holds out 2–3 structures for validation while training on the remainder.

Training hyperparameters: Adam optimiser [20] with learning rate 3×10^{-4} and weight decay 10^{-4}; cosine annealing schedule; gradient clipping at $\|\nabla\|_{\max} = 1$; early stopping on validation loss with patience 25 and maximum 150 epochs; batch size 16.

We report accuracy, area under the ROC curve (AUC), and per-class accuracy as fold means ± standard deviation across five folds.

2.7 Pair distribution function analysis

To test whether standard structural characterisation can distinguish the two cooling rates, we compute partial pair distribution functions $g_{\alpha\beta}(r)$ for all three pair types (α–β $\in$ {Si–O, Si–Si, O–O}) from each full 3000-atom structure with 200 radial bins up to $r_{\max} = 8.0$ Å. We apply two-sample Kolmogorov–Smirnov (KS) tests between the fast (100 K/ps) and slow (0.1 K/ps) $g(r)$ values, pooled across six samples per class. Significance threshold: $\alpha_{\mathrm{KS}} = 0.05$.

3 Results

3.1 Ablation: the geometric carrier is system-dependent

Table 2 and Fig. 1 present the central result: systematic ablation across four feature sets on SiO_2, compared with prior results on pair-potential glasses from Ref. [5].

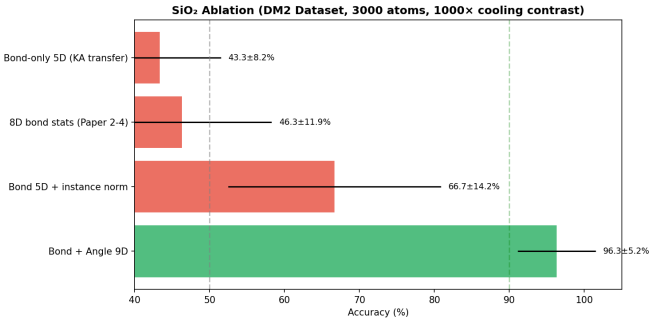


Figure 1: Ablation hierarchy for SiO_2 thermal history classification. DM2 dataset, 3000-atom structures, 1000× cooling rate contrast, five-fold StratifiedGroupKFold CV. Bond-only features (5D and 8D) produce near-chance accuracy (red). Bond + Angle (9D) achieves 96.3% (green). Error bars: $\pm 1\sigma$ across folds.

The results reveal a sharp contrast between glass families:

Bond-length features fail in SiO_2. The 5D bond-only features that achieve 95.10 ± 1.02% in LJ and 96.23 ± 0.74% in KA produce 43.3 ± 8.2% in SiO_2—consistent with the model collapsing to a single-class prediction on each fold. The 8D extended features perform similarly (46.3 ± 11.9%). Bond-length statistics carry no discriminative signal for thermal history in the tetrahedral network.

Angular geometry is the carrier. Adding four bond-angle features (O–Si–O mean and standard deviation, Si–O–Si bridging angle) to the bond statistics restores accuracy to 96.3 ± 5.2% (AUC = 0.981). Per-class accuracy is balanced at 96.0% for both fast and slow glasses. Per-fold results range from 86.7% to 100.0%, with three of five folds achieving perfect classification.

Instance normalisation yields a partial signal. Bond features with instance normalisation achieve 66.7 ± 14.2%, above chance but with high variance (fold-level range 43.3%–85.0%). Removing global magnitude information exposes a weak geometric signal from the relational structure of the bond network, but insufficient for reliable classification.

3.2 Pair distribution functions: standard characterisation is blind

Figure 2 compares the partial $g(r)$ for the fastest (100 K/ps) and slowest (0.1 K/ps) glasses. All three partial PDFs are visually indistinguishable across the full range $0 < r < 8$ Å. KS tests confirm no statistically significant difference:

- Si–O: KS = 0.050, $p = 0.10$
- Si–Si: KS = 0.041, $p = 0.27$
- O–O: KS = 0.043, $p = 0.21$

All p-values exceed the $\alpha_{\mathrm{KS}} = 0.05$ threshold. The pair distribution function—the primary structural observable from neutron and X-ray total scattering—cannot distin-

Table 2: Classification accuracy across glass families and feature sets. SiO_2 results: five-fold StratifiedGroupKFold CV on the DM2 dataset [16] (10 patches × 512 atoms per structure, 120 graphs total). LJ and KA results from Ref. [5]. All use the identical GATv2 architecture. '—': not tested for that system.

Feature set	Dim	LJ ($N = 256$)	KA ($N_{\text{sub}} = 512$)	SiO_2 ($N_{\text{sub}} = 512$)	Interpretation
Bond-only (5D)	5	95.10 ± 1.02%	96.23 ± 0.74%	43.3 ± 8.2%	Bond lengths: no signal in SiO_2
Extended bond (8D)	8	—	—	46.3 ± 11.9%	Additional bond stats: still no signa
Bond + inst. norm (5D)	5	94.5%	—	66.7 ± 14.2%	Weak relational signal only
Bond + Angle (9D)	**9**	—	—	**96.3 ± 5.2%**	**Angular geometry is the carrie**

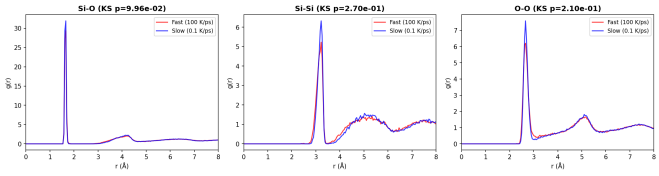


Figure 2: Partial pair distribution functions $g(r)$ for fast (100 K/ps, red) and slow (0.1 K/ps, blue) SiO_2 glasses. All three pair types are visually indistinguishable. KS tests confirm no significant difference ($p > 0.05$), establishing that $g(r)$ is blind to the thermal history signal detected by the GNN at 96.3%.

guish glasses whose thermal histories differ by three orders of magnitude. The GNN with angular features classifies the same glasses at 96.3%. The dominant carrier of thermal memory in silica is therefore a many-body angular correlation that projects negligibly onto the pair distribution function.

3.3 Latent space structure

Figure 3 shows the PCA projection of the 64-dimensional GATv2 latent vectors from the Bond + Angle model. PC1 captures 41.3% of variance and PC2 captures 16.6%. The Spearman correlation between PC1 and the cooling-rate label is $\rho = -0.135$ ($p = 0.14$), indicating that the classification signal is distributed across multiple latent dimensions rather than concentrated on a single axis.

The latent space shows visible clustering by parent structure: patches from the same 3000-atom glass map to tight groups at the periphery. This is expected since patches from the same glass share thermal history and similar medium-range structure. The inter-cluster separation across multiple PCs enables the high classification accuracy.

This contrasts with the LJ system, where PC1 captured 76.9% of variance with strong monotonic ordering [5], suggesting that the angular encoding of thermal history in network glasses is geometrically more complex than the bond-length encoding in pair-potential glasses.

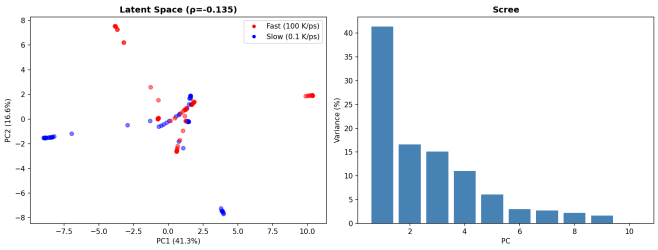


Figure 3: Left: PCA of the 64-D GATv2 latent space (Bond + Angle model). Red: fast (100 K/ps); blue: slow (0.1 K/ps). Patches from the same parent structure cluster together. Right: scree plot showing distributed variance (41.3%, 16.6%, 15.0%, ...), contrasting with the single-dominant-PC structure in LJ [5].

4 Discussion

4.1 The geometric carrier is system-dependent

The central finding is that the geometric degree of freedom encoding thermal history depends on the bonding topology of the glass former. In pair-potential glasses (LJ, KA), the carrier is bond-length statistics [5]. In SiO_2, bond lengths carry no signal; the carrier is the angular geometry of the network.

This can be understood from the physics of each system. In pair-potential glasses, atoms interact isotropically and local structure is characterised by packing efficiency, which manifests as variations in nearest-neighbour distances. In SiO_2, the covalent Si–O bond length is tightly constrained at ∼1.62 Å with minimal variation between glasses. However, the network topology—how tetrahedra connect through bridging oxygens—admits substantial angular flexibility. The Si–O–Si bridging angle distribution spans ∼130°–180° [12, 11], and its width and shape depend on thermal history [12]. Our ablation quantitatively confirms that this angular degree of freedom is the structural repository of cooling-rate memory.

4.2 Directional consistency with experimental thermal-history signatures

While our classification results are derived entirely from simulated glasses, it is important to ask whether the angular carrier identified by the GNN corresponds to a signal

that is also observable in real, laboratory-produced silica. Independent evidence from three experimental lineages, spanning more than two decades, supports the qualitative direction of our finding.

Raman spectroscopy of vitreous silica shows that spectral features assigned to the continuous random network shift systematically with fictive temperature, in a manner attributed to a contraction of the average Si–O–Si angle as fictive temperature (and, by extension, cooling rate) increases [13]. Independently, infrared spectroscopic studies of the $\sim$1122 cm^{-1} and $\sim$2260 cm^{-1} silica structural bands show that the peak wavenumber decreases with increasing fictive temperature, a shift that has been quantitatively linked to a decreasing average Si–O–Si bond angle via a central- and non-central-force network dynamics model [14, 15]. In each case, the reported direction is the same: faster effective cooling (higher fictive temperature) drives the Si–O–Si angle distribution toward smaller, more strained values, while the Si–O bond length itself is not reported as a useful discriminator of thermal history.

This directional trend is also reproduced in an independent molecular dynamics study using the BKS potential [12], which found the mean Si–O–Si angle to shift from 141° at the fastest cooling rate to 152° at the slowest cooling rate simulated, while the Si–O bond length remained essentially unchanged across the same range of cooling rates. This is a fully independent computational lineage from the present work: different interatomic potential (BKS vs. the Jakse potential used here), different methodology (direct bond-angle histograms rather than a learned graph-attention classifier), and a different research group. The qualitative agreement across four independent sources—two spectroscopic, one simulation-based, and the present GNN-based study—that angular geometry, not bond length, is the structural quantity most sensitive to thermal history in SiO_2 strengthens the physical interpretation of our ablation result.

We emphasise that this is a validation of *direction and mechanism only*. The magnitude of the angular shift observed in hyper-quenched MD simulations (cooling rates of 10^{12}–10^{14} K/s) is not directly comparable to the much smaller shifts reported in laboratory-cooled glasses (cooling rates of 10^{0}–10^{5} K/s), a well-documented consequence of the many-orders-of-magnitude gap between simulated and experimental cooling rates [12]. A quantitative reconciliation of these magnitudes, for example via the extrapolation approaches developed for sodium silicate glasses, is left to future work.

4.3 Implications for structural characterisation

The $g(r)$ blindness result has practical implications. Neutron and X-ray total scattering experiments access the pair distribution function via the structure factor $S(Q)$. Our analysis demonstrates that $g(r)$ cannot distinguish SiO_2 glasses whose cooling rates differ by 1000×—yet the GNN classifies them at 96.3% using angular information.

This means the dominant structural carrier of thermal memory in silica is a many-body angular correlation that projects negligibly onto pair correlations. This is consistent with theoretical expectations: $g(r)$ captures two-body correlations, while bond angles are inherently three-body quantities. However, the *quantitative magnitude* of this gap—from complete blindness ($p > 0.05$) to near-perfect discrimination (96.3%)—has not been established previously for a specific glass system using controlled ablation.

Techniques sensitive to angular correlations include: three-body correlation analysis of total scattering data [22], Raman spectroscopy probing ring-breathing modes [23], and NMR spectroscopy sensitive to local coordination environments. Our results provide quantitative motivation for preferring these over standard $g(r)$ analysis when characterising the thermal state of silicate glasses.

4.4 Architecture transferability

The GATv2 architecture required zero modification to transfer from pair-potential to network glasses. The same two-layer, four-head architecture achieves >95% accuracy across three fundamentally different glass families, provided appropriate input features. What changes is which features carry the signal, not the architecture.

This supports using GATv2 as a general-purpose structural probe for disordered materials. The ablation methodology provides interpretable, system-specific insight into which geometric degrees of freedom encode processing history.

4.5 Limitations

Small dataset. Only 6 structures per cooling rate (12 for binary). Patch extraction increases effective sample size to 120 graphs, and group CV prevents leakage, but statistical power is limited. The 5.2% fold-to-fold standard deviation reflects this.

Binary classification only. We test only the extreme rates (0.1 vs. 100 K/ps). Multi-class classification or regression across all four rates requires more data.

Single potential. The Jakse potential [17] is well-validated but results may differ with BKS [21], ReaxFF, or ML potentials.

Angle-only experiment not performed. We tested Bond + Angle (9D) but not angle-only (4D), which would determine whether bond-angle information alone suffices.

Patch size. 512-atom patches may not capture the full extent of medium-range order in SiO_2 [11]; larger patches could reveal additional signatures.

Quantitative sim-to-experiment gap not closed. We show directional agreement between our simulated angular carrier and experimental Si–O–Si angle trends from Raman and IR spectroscopy (Sec. 4.2), but a quantita-

tive reconciliation across the ∼8–12 order-of-magnitude gap between MD and laboratory cooling rates has not been attempted here.

5 Conclusions

We have demonstrated that the geometric carrier of thermal history is system-dependent:

(1) In pair-potential glasses (LJ, KA), bond-length statistics achieve >95% classification accuracy.

(2) In SiO_2, bond-length statistics carry no signal (43–46%); adding bond angles restores accuracy to $96.3 \pm 5.2\%$.

(3) The pair distribution function $g(r)$ cannot distinguish SiO_2 glasses differing by 1000× in cooling rate ($p > 0.05$ for all partial PDFs).

These findings establish that geometric encoding of thermal history is present in all tested glass families, but the specific carrier depends on bonding topology: bond lengths in isotropic pair-potential systems, bond angles in directional network glasses. The practical implication is that characterisation techniques sensitive to angular correlations may be more informative than standard $g(r)$ analysis for assessing the thermal state of silicate glasses.

The GATv2 architecture transfers without modification across glass families, achieving >95% in each, supporting its use as a general structural probe for disordered materials.

Data and Code Availability

SiO_2 configurations: DM2 dataset [16], https://github.com/digital-synthesis-lab/DM2. GNN code: https://github.com/Saurabsingh778/sio2-angular-thermal-memory.

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